

Project 1: Optimal Control

Controlled dynamical system (finite-horizon)

$$y'(t) = g(y(t), a(t)) \quad \text{for all } t \in [0, T]$$

$$\therefore \min_a \int_0^T c(y(t), a(t)) dt + c_{term}(y(T))$$

(or expectation...)

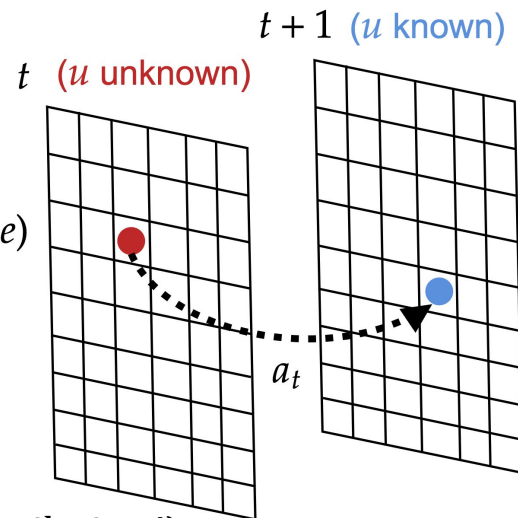
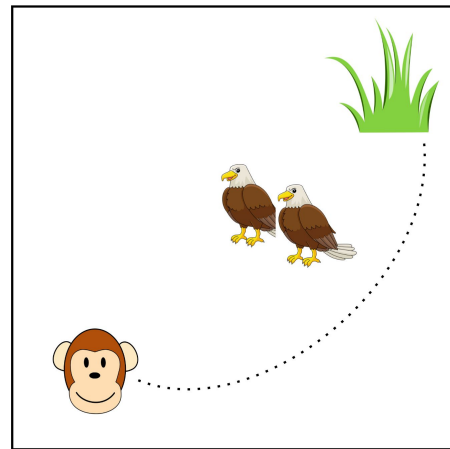
Value func via
dynamic
programming:

$$u(y, t) = \min_{a \in \mathcal{A}} \underbrace{\{u(y + dy, t + dt)\}}_{\text{land here}} + \underbrace{c(y, a) \cdot dt}_{\text{running cost}} \quad (x, y, e)$$

HJB PDE:

$$u_t + \min_{a \in \mathcal{A}} \{u_y \cdot g(y, a) + c(y, a)\} = 0$$

Sweep opposite to direction of “causality” (not necessarily time!)



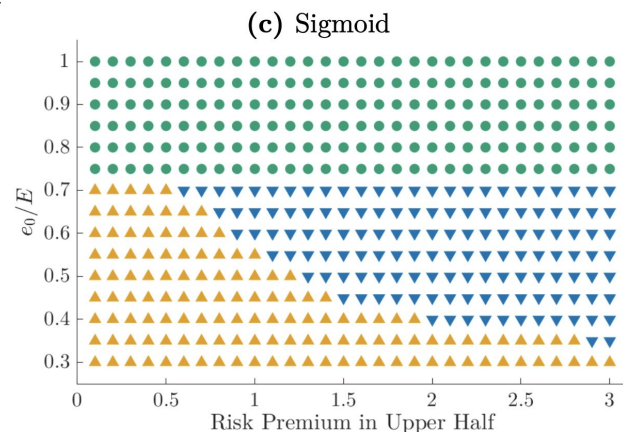
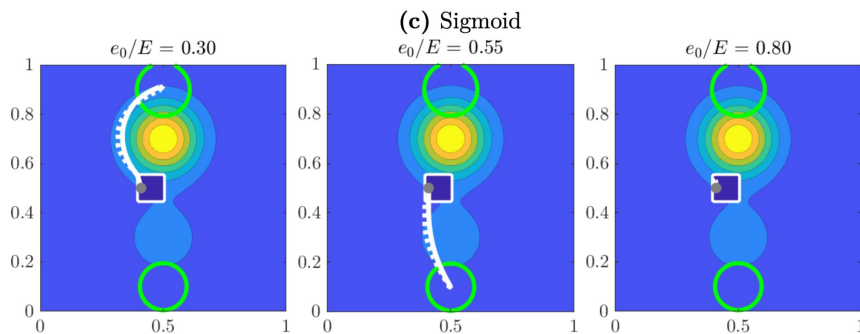
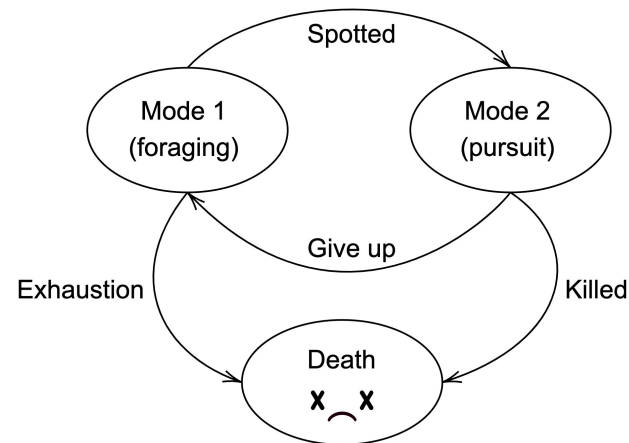
Foraging Model

More realistic model of prey foraging behavior

State: position in the plane and energy

Goal: terminal utility $U(e_T)$ & return home

Randomness: the prey being spotted by predator! $\rightarrow$ PDMP



Paths under sigmoid utility. Source: Working Paper

Extensions/Broader Relevance

- Prey could have imperfect knowledge of risk terrain
→ Bayesian reinforcement learning?
- Both prey *and* predators have agency → differential game
- Dynamic programming and curse of dimensionality:
neural optimal control, or neural HJB PDE solvers?

Project 2: Percolation Theory

Micro interactions $\rightarrow$ macro phase transitions

Simplest: Bernoulli bond percolation on $\mathbb{Z}^d$

Study “percolation probability” $\theta(p)$

Equals **0** near 0, **+** near 1; by coupling, increasing

What is behavior at/near critical point p_c ?

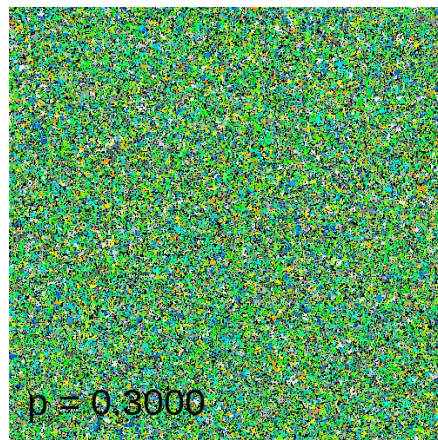
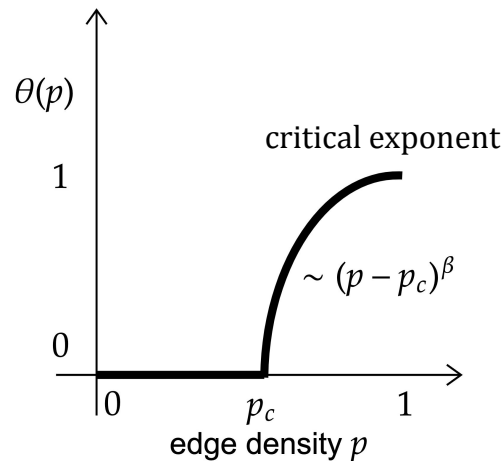
(Continuous? Exponents? *Sharpness*?)

Key: *differential inequalities*. (doesn't solve everything...)

Aizenman-Barsky '87: $\frac{\partial M}{\partial \beta} \leq 2dJ \cdot M \frac{\partial M}{\partial h}$ Burgers' ineq
(M defined on **next** slide...)

$$\longrightarrow M \leq h \frac{\partial M}{\partial h} + \beta M \frac{\partial M}{\partial \beta} + M^2 \quad \Phi^3\text{-ineq}$$

($p = 1 - e^{-\beta J}$)



Emergence of ∞ cluster.
Source: Wikipedia

DRT inequality and Φ^3 -inequality

$$\text{RCM: } \frac{1}{Z} \underbrace{q^{N_C}}_{\text{weighting}} \prod_{e \in E} p^{\omega_e} (1-p)^{1-\omega_e}$$

Goal: generalize Φ^3 -ineq to (dependent) *Random Cluster Model*.

(To start,) f increasing function of *independent* boolean inputs.

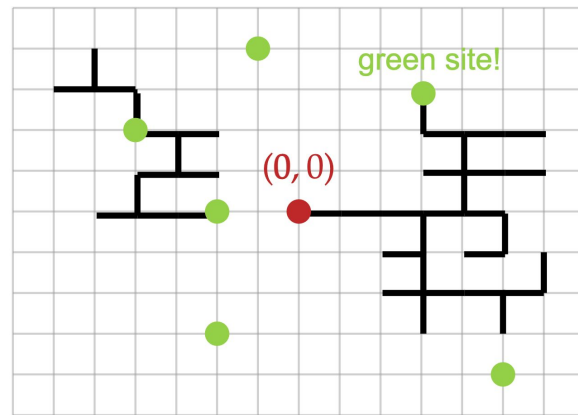
Discrete Poincaré: $\text{Var}[f(\omega)] \leq \sum_{e \in E} \underbrace{\text{Cov}[f(\omega), \omega_e]}_{\text{influence of } e} \left(= p(1-p) \cdot \frac{d}{dp} E[f(\omega)] \right)$
 $\rightarrow$ S-shape of $\theta_n(p)$

OSSS ('05): $\text{Var}[f(\omega)] \leq \sum_{e \in E} \underbrace{\delta(T, e)}_{\text{revelment prob}} \cdot \underbrace{\text{Cov}[f(\omega), \omega_e]}_{\text{influence}}$

T = decision tree computing f

DRT ('19): *same* holds for $\omega \sim$ monotone μ

App (Aizenman-El Bahri): T computing indicator f of $\{0 \leftrightarrow \text{green sites}\} \rightarrow \Phi^3\text{-ineq} \blacksquare$



$\mathcal{G}$: green sites

$M := P(0 \leftrightarrow \mathcal{G})$

Extensions/Broader Relevance

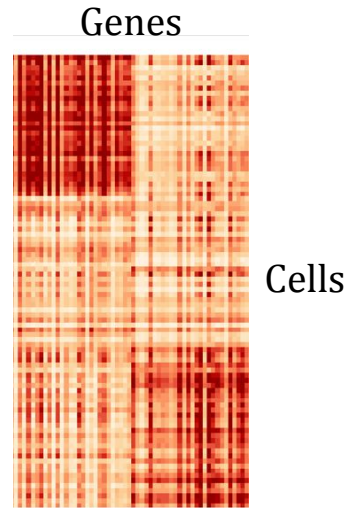
- Generalizing the Burgers' inequality requires new ideas
- Percolation/Ising models can be defined on arbitrary graphs
- Recent applications to statistical inference on graphs

Project 3: Density-Difference Estimation

Differential Abundance Problem (DAP). Let f_1 and f_0 be (unknown) densities supported on a manifold $\mathcal{M}$. Let $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ and $\mathbf{X}' = \{\mathbf{x}'_1, \dots, \mathbf{x}'_{N'}\}$ be datasets sampled i.i.d. from f_0 and f_1 , respectively. Determine the subset of $\mathcal{M}$ on which $f_1 > f_0$, i.e. the set

$$\{x \in \mathcal{M} : f_1(x) > f_0(x)\}$$

Motivation: scRNA-seq



One approach: kernel density estimation (KDE).

LSDD (Sugiyama et al.): estimate density-diff *directly*; **Euclidean** case

Source: Raphael Group

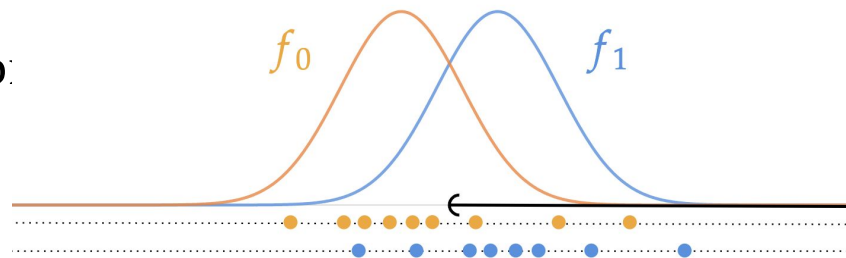
$$g(x) = \sum_{i=1}^{N+N'} \theta_i \exp\left(-\frac{\|x - y_i\|^2}{2\sigma^2}\right) \quad \text{via LS (must plug-in): } \operatorname{argmin}_{\theta \in \mathbb{R}^{N+N'}} \int_{\mathbb{R}^G} (g(x) - (f_1 - f_0)(x))^2 dx + \lambda \|\theta\|^2$$

But $\mathcal{M}$ is often non-Euclidean. Generalize?

OurMethod: construct $\{w_i\}$; consider estimator:

$$g(x) = \sum_{i=1}^{N+N'} \theta_i w_i(x)$$

Random walk pmf
on heat kernel graph
with cutoff



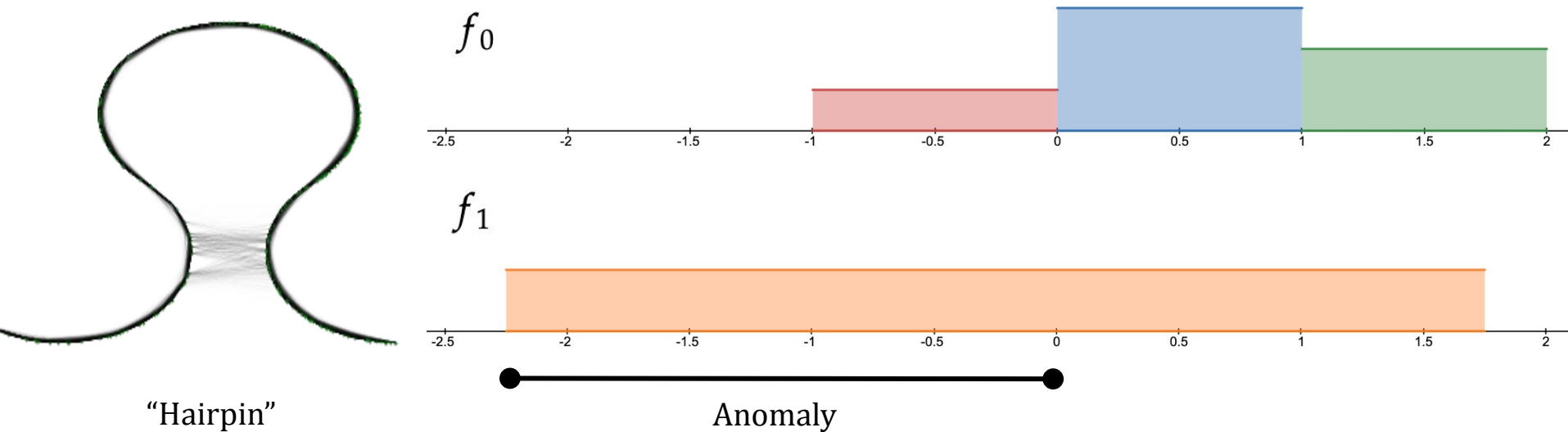
Synthetic example

Construct two piecewise-constant densities along certain curve.

OurMethod F-score: **0.91** averaged over 200 iterations.

LSDD F-score: 0.81. (We seem to make less errors near bottleneck; “captured geometry”.)

($n=300$ samples, each w.p. 0.5, random walk pmf with $k=20$, $t=150$.)



Extensions/Broader Relevance

- Best choice of weighted graph? (Still experimenting)
- Laplacian regularization? (But MELD paper does this...)
- Generally: how to develop automated procedures for selecting parameters? (Ex: heat kernel variance, # of timesteps) (Manifold learning seems very sensitive to parameter choices...)
- “Classical” methods vs neural approaches (GANs)?

Favorite Coursework

My three favorite courses from undergrad.

The ideas we covered seem to have wide applicability.

- MAT 325: Fourier Series and Partial Differential Equations
- ORF 550: Probability in High Dimension
- ORF 524: Statistical Theory and Methods